$$= IC \times (1.5 \times 20\,us + \overbrace{1.3 \times 0.039 \times 200\,us}^{10.14\,us}) =$$

$$= IC \times \boxed{40.14\,us}$$

1-way is better!

average time per instruction (taking into account the cache impact)

$$CPU_{time\ 2\text{-way}} = IC \times (1.5 + 1.3 \times 0.03 \times \text{Miss penalty in cc}) \times \underbrace{\text{Clock cycle time}}_{20\,us}$$

$$= IC \times (1.5 \times 21.7_{cc} + 1.3 \times 0.03 \times 200\,us)$$

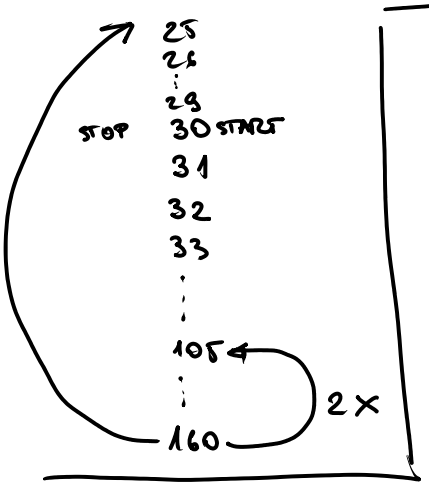
$$= IC \times \boxed{40.35\,us}$$

Example 3

- A cache 4-way SA has 8 sets and 4 words/block
- MM size is $\boxed{2^{16}\text{ words}}$; the addressable unit is the word
- The cache is initially empty
- t_{ac} = 30 us = Clock cycle time
- Miss Penalty = 480 us
- LRU replacement policy

a) Memory address format, total cache capacity if the word is on 32 bits.

b) Calculate the Hit Ratio and AMAT for the following workload



a)

15 … 5	4 3	2 1 0
Tag	Index	Block offset
<11>	<3>	<2>

V	Tag	Age	Data
<1>	<11>	<2>	4B/word × 4 words/block = 2^4B/block

$$\text{Cache capacity} = \underbrace{2^3}_{\text{sets}} \times \underbrace{2^2}_{\text{S.A}} \times (14\text{ bits} + 2^4\text{B}) =$$

$$2^6 = \frac{2^{10}}{2^4}$$

$$\cong \underbrace{2^5 \times 2^4\text{B}}_{\text{data}} + \underbrace{2^5 \cdot 2^1\text{B}}_{2^6}$$

$$\cong 0.5\text{ KiB} + 0.0625\text{ KiB}$$

$$\cong 0.5625\text{ KiB}$$

b) TATA

Address	Block No
0 - 3	0
4 - 7	1
8 - 11	2
12 - 15	3
16 - 19	4
20 - 23	5
24 - 27	6
28 - 31	7
32 - 35	8
36 - 39	9
40 - 43	10
44 - 47	11
48 - 51	12
52 - 55	13
56 - 59	14
60 - 63	15
64 - 67	16
68 - 71	17
72 - 75	18
76 - 79	19
80 - 83	20
84 - 87	21
88 - 91	22
92 - 95	23
96 - 99	24
100 - 103	25
104 - 107	26
108 - 111	27
112 - 115	28
116 - 119	29
120 - 123	30
124 - 127	31
128 - 131	32
132 - 135	33
136 - 139	34
140 - 143	35
144 - 147	36
148 - 151	37
152 - 155	38
156 - 159	39
160 - 163	40
164 - 167	41
168 - 171	42
⋮	⋮

≙ 0.5 KiB + 0.0625 KiB

≙ 0.5625 KiB

Bank 0

	V	Tag	Data
0	1	1	~~32 - 35~~
1	1	1	36 - 39
2	1	1	40 - 43
3	1	1	44 - 47
4	1	1	48 - 51
5	1	1	52 - 55
6	1	1	~~56 - 59~~
7	1	1	~~28 - 31~~
4			156 - 159

Bank 1

	V	T	Data
0	1	2	64 - 67
1	1	2	68 - 71
2	1	2	72 - 75
3	1	2	76 - 79
4	1	2	80 - 83
5	1	2	84 - 87
6	1	2	88 - 91
7	1	~~1~~ 0	~~60 - 63~~
			28 - 31

Bank 2

	V	T	Data
0	1	3	96 - 99
1	1	3	100 - 103
2	1	3	104 - 107
3	1	3	108 - 111
4	1	3	112 - 115
5	1	3	116 - 119
6	1	3	120 - 123
7	1	2	92 - 95

Bank 3

	V	T	Data
0	1	4	128 - 131
1	1	4	132 - 135
2	1	4	136 - 139
3	1	4	140 - 143
4	1	4	144 - 147
5	1	4	148 - 151
6	1	4	152 - 155
7	1	3	124 - 127

$$\begin{array}{l} 1TM \\ 1H \end{array} \left(\begin{array}{l} 1TM \\ 3H \end{array}\right) \times 31 \left(\begin{array}{l} 1TM \\ 3H \end{array}\right) \left(\begin{array}{l} 1TM \\ 0H \end{array}\right) \left(\begin{array}{l} 0TM \\ 56H \end{array}\right) \times 2$$

$$\left(\begin{array}{l} 1TM \\ 2H \end{array}\right) \left(\begin{array}{l} 1TM \\ 2H \end{array}\right)$$

$39 \,|\, 8$, $32 \,|\, 4$, 7

$40 \,|\, 8$, 5

$19 \,|\, 8$, $16 \,|\, 2$, 3

$$\begin{array}{l} 32TM \\ 96H \end{array} \rightarrow \begin{array}{l} 33TM \\ 97H \end{array} \rightarrow \begin{array}{l} 34TM \\ 97H \\ \hline 112 \end{array} \rightarrow \begin{array}{l} 34TM \\ 209H \end{array} \rightarrow$$

$$\rightarrow \begin{array}{l} 36TM \\ 213H \\ \hline 249 \text{ accesses} \end{array}$$

$$\text{Hit Rate} = \frac{213}{249} = 85.54\%$$

$$\text{Miss Rate} = 14.46\%$$

$$ATMAT = 30\,ns + 14.46\% \times 480\,ns =$$

$$= \boxed{99.408\,ns}$$